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HUMAN RESOURCE MANAGEMENT | RESEARCH ARTICLE

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Strategic human resource management in the digital economy era: an empirical study of challenges and opportunities among MSMEs and startups in Indonesia

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ABSTRACT

This study investigates the impact of digital Human Resource Management (HRM) strategies on workforce productivity and organizational performance within Micro, Small, and Medium Enterprises (MSMEs) and startups in Indonesia. Anchored in the Resource-Based View (RBV) and Dynamic Capabilities Theory, the research conceptualizes digital HRM as a strategic asset that enhances organizational performance by enabling firms to develop, deploy, and reconfigure human capital capabilities in response to digital transformation. The conceptual model includes five constructs: Digital Technology Adoption in HRM, Digital Training and Development, Work Flexibility and Hybrid Management, Innovation in Recruitment and Retention, and Utilization of HR Analytics and Artificial Intelligence (AI). Data were collected through structured questionnaires distributed to MSME and startup leaders and analyzed using Structural Equation Modeling (SEM) with Smart PLS. The results indicate that Digital Technology Adoption, Digital Training and Development, and Work Flexibility do not significantly influence workforce productivity or organizational performance ($p > 0.05$). However, Innovation in Recruitment and Retention and the Utilization of HR Analytics and AI have a significant and positive impact on both workforce productivity and performance outcomes ($p = 0.000$). These findings highlight that while not all digital HR strategies yield immediate benefits, the ability to build dynamic capabilities through innovative talent practices and data-driven HR systems is essential to achieving competitive advantage and organizational success in the digital economy.

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Business, Management and Accounting; Economics; Education Studies; Human Resource Management

1. Introduction

The digital economy has rapidly transformed global business, compelling organizations to adopt innovative strategies to stay competitive. The integration of digital technologies has reshaped traditional business models, requiring greater agility, adaptability, and workforce efficiency (Tarique & Schuler, 2018). This shift has also revolutionized Human Resource Management (HRM), positioning Strategic Human Resource Management (SHRM) as a critical driver of success in aligning workforce capabilities with technological advancements (Bondarouk & Brewster, 2016). In local enterprises, Chief among these are digital skill gaps, resistance to change, and resource constraints. Employees often lack training in digital tools, and organizational readiness is hampered by fears of job displacement and insufficient investment in digital HR infrastructure (Frey & Osborne, 2017; Vakola, 2016). Despite these barriers, SHRM offers pathways to improved productivity and competitiveness. Digital training enhances adaptability, flexible work arrangements boost efficiency and satisfaction, and AI-driven HR tools enable more effective decision-making (Chamorro-Premuzic et al., 2016; Kossek & Thompson, 2016). Innovations in recruitment and analytics further support strategic workforce management (Stone & Deadrick, 2015).

This study explores how SHRM can enhance local enterprise competitiveness by fostering workforce adaptability and organizational agility (Sparrow et al., 2016). It investigates the factors that shape

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effective SHRM implementation—digital adoption, training, and leadership—and examines how technology integration, employee development, and performance outcomes interconnect (Bondarouk & Brewster, 2016).

In Indonesia, MSMEs face distinct challenges in the digital shift, including limited digital literacy and inadequate infrastructure (Prasetio et al., 2023; Said & Soi, 2025). Reliable internet access remains uneven, while financial and regulatory constraints further complicate transformation efforts (Farliana et al., 2023; Fridayani & Chiang, 2022). Nonetheless, digital tools—such as e-commerce, digital marketing, and cloud solutions—offer clear benefits for efficiency and competitiveness (Rahayu et al., 2024; Soekiman, 2024). To realize these benefits, MSMEs must develop digital skills and HR capacity, with SHRM playing a vital role in workforce development and performance enhancement (Soekiman, 2024). Government support, private sector collaboration, and international cooperation are key to improving infrastructure and enabling innovation (Fridayani & Chiang, 2022). SHRM's focus on hard and soft skills, including digital marketing and tech integration, supports sustainable enterprise growth in the digital era.

The primary aim of this research is to analyze how SHRM supports workforce development, HR optimization, and business performance in local enterprises (Stone & Deadrick, 2015). It identifies practice gaps and offers recommendations for improving digital HRM strategies (Thite, 2022), and furthermore, highlighting both challenges and opportunities (Jiang & Messersmith, 2018). By examining digital workforce readiness, the study provides evidence-based insights for enhancing skills, integrating HR tech, and fostering talent strategies (Priyono et al., 2020).

Recent work highlights SHRM's value in addressing major societal and economic challenges. Even in resource-limited, non-Western contexts, SHRM helps deliver public value and navigate crises (Mousa et al., 2025; Mousa & Arslan, 2025a). It acts as a 'lifebuoy' for employee well-being and organizational continuity during economic shocks (Mousa & Arslan, 2025b). These insights are relevant to MSMEs, which must remain agile amid fast-evolving digital pressures. Although SHRM's role in digital transformation is well-theorized, research on its application in Indonesian MSMEs remains limited. These enterprises often lack digital literacy, infrastructure, and strategic foresight (Alfarizi et al., 2024; Hasbullah et al., 2024). Tailored SHRM approaches are essential, addressing unique constraints while enabling digital innovation and competitiveness (Anatan & Nur, 2023). Emerging frameworks emphasize strategic alignment, tech selection, and partnerships for knowledge transfer (Ananda et al., 2023; Prihandono et al., 2024).

This study addresses the empirical gap by examining which SHRM components most effectively support MSMEs in digital transformation. It applies theoretical lenses such as the Resource-Based View and Dynamic Capabilities Theory to explore how SHRM enhances innovation, agility, and sustainable advantage under constrained conditions. Focusing on practices like digital recruitment, HR analytics, training innovation, and hybrid work management, this research offers theory-informed, practical insights for Indonesian MSMEs in the digital era.

2. Literature review

2.1. Strategic human resource management (SHRM) concept

Strategic Human Resource Management (SHRM) is an advanced approach to managing human resources, aligning HR practices with organizational goals to gain a sustainable competitive advantage. According to Wright and McMahan (2011), SHRM focuses on linking HR policies with the overall business strategy to enhance organizational performance. This approach moves beyond traditional HR functions by integrating workforce planning, talent management, and performance measurement into the strategic decision-making process (Becker & Huselid, 2006). The concept of SHRM is grounded in several theoretical perspectives, including the Resource-Based View (RBV) and the Human Capital Theory. The RBV, as proposed by Barney (1991), suggests that human resources can serve as a source of competitive advantage if they are valuable, rare, inimitable, and non-substitutable. Similarly, Human Capital Theory, articulated by Becker (2009), emphasizes the importance of investing in employee development to enhance skills, productivity, and innovation, ultimately driving business success (Boxall & Purcell, 2022).

A key aspect of SHRM is strategic fit, which refers to aligning HRM practices with both external business environments and internal organizational needs. Delery and Doty (1996) introduced universalistic,

contingency, and configurational perspectives to explain how HR practices can contribute to performance. The universalistic approach posits that certain HR practices (e.g. high-performance work systems) consistently lead to better outcomes, while the contingency perspective argues that HR strategies must align with the organization's strategic orientation. The configurational approach, on the other hand, emphasizes the synergy between multiple HR practices that collectively impact performance (Lengnick-Hall et al., 2011). Research on SHRM also highlights the role of strategic workforce planning and HR analytics in enhancing decision-making processes. With advancements in HR technology and data-driven decision-making, organizations now leverage HR analytics to assess workforce trends, predict future talent needs, and improve employee engagement (Marler & Boudreau, 2017). These strategic HR initiatives contribute to innovation, agility, and long-term business sustainability.

The implementation of SHRM is influenced by various external and internal factors, including globalization, technological advancements, and changing workforce demographics. For example, in response to increasing digitalization, organizations are adopting AI-driven talent management strategies to enhance recruitment and retention processes (Stone et al., 2015). Furthermore, globalization has necessitated cross-cultural HRM approaches to manage diverse and geographically dispersed workforces effectively (P. Sparrow et al., 2016). Despite its numerous advantages, SHRM also faces challenges, such as resistance to change, difficulties in measuring HR effectiveness, and alignment issues between HR and top management. Scholars suggest that for SHRM to be successful, organizations must foster a culture of collaboration, continuous learning, and adaptability (Ulrich et al., 2012). Additionally, HR professionals must develop strategic competencies to effectively contribute to business outcomes (Schuler & E. Jackson, 2014).

In this study, the digital economy is interpreted as an ecosystem driven by the integration of automation, artificial intelligence (AI), digital platforms, and e-commerce models into various business operations. This transformation reshapes how Micro, Small, and Medium Enterprises (MSMEs) operate, engage with customers, and manage human resources. However, digital adoption among MSMEs in Indonesia still faces significant challenges, including limited infrastructure, low digital literacy, and financial constraints. The perspective of Strategic Human Resource Management (SHRM), digital transformation demands a more proactive role from HR in managing these changes. The use of AI and automation in HR processes, such as recruitment, performance management, and employee engagement, offers increased efficiency and data-driven decision-making (Johnson et al., 2021). Yet, the effective implementation of these technologies requires not only digital competence among employees but also cultural readiness within organizations (Iskandar, 2023). Moreover, digital business models such as e-commerce and digital platforms are reshaping the workforce structure in MSMEs, leading to an increased demand for remote work capabilities and workforce flexibility. This necessitates HR strategies that support flexible work arrangements while maintaining high levels of employee commitment and well-being. For overcome these challenges, collaborations between MSMEs and educational institutions are crucial for effective knowledge transfer and enhancing digital readiness. Structured training programs focused on digital skills development can accelerate digital transformation processes and significantly improve business competitiveness (Anatan & Nur, 2023).

The adoption of Strategic Human Resource Management (SHRM) practices among Indonesian Micro, Small, and Medium Enterprises (MSMEs) and startups is increasingly recognized as a critical factor in enhancing competitiveness, agility, and long-term sustainability in the digital economy. A growing body of empirical evidence demonstrates that SHRM, when properly integrated with digital tools and strategic frameworks, can serve as a catalyst for organizational transformation and performance improvement. A quantitative study of MSMEs in Jakarta's foodservice sector found that nine out of ten core HRM practices, including workforce planning, talent acquisition, training and development, performance evaluations, and career progression, have a significant positive impact on business performance. This emphasizes the importance of a holistic SHRM framework in aligning human capital capabilities with organizational goals, particularly in labor-intensive service industries (Nurianna et al., 2023). In Bandung, a small frozen food enterprise with under ten employees illustrates how digital HRM tools, such as WhatsApp for sales and omni-channel integration for inventory and performance monitoring, enable lean organizations to scale and remain competitive. These practices show how SHRM can support process digitalization and employee coordination in micro-sized enterprises through low-cost, adaptable technologies (Djamaludin et al., 2024). Similarly, informal retail shops, known as '*Warung*', are undergoing digital HRM transitions

through partnerships with tech startups. These collaborations integrate information and communication technology (ICT) into business models, focusing on employee readiness, digital upskilling, and organizational adaptability. Identifying success factors such as clear HR governance structures and workforce engagement is essential for ensuring scalability and ownership transferability (Muhtadi & Ardi, 2021). In rural regions such as Ogan Ilir Regency, government-led HRM interventions—though still limited have facilitated growth in culinary MSMEs through financial support, HR training, and marketing assistance. These efforts underscore the role of SHRM in regional economic development and in positioning local enterprises within the broader creative and tourism economy (Semil et al., 2020). Furthermore, a comprehensive study of handicraft MSMEs at the Indonesia–Malaysia border reveals that SHRM, when aligned with entrepreneurial capabilities and creativity (as theorized by the Theory of Planned Behavior), substantially enhances SME sustainability and employee innovativeness (Asriati et al., 2022). Strategic HRM in this context empowers business actors not only to survive but to thrive amidst cross-border trade dynamics and cultural diversification. For high-growth startups, research highlights the implementation of Management Control Systems (MCS) embedded within HR structures as instrumental in fostering radical innovation. External market pressures and organizational culture influence these systems, which, in turn, enable startups to cultivate collaborative environments, knowledge sharing, and agile response mechanisms to dynamic market conditions (Piliang et al., 2025). These multi-sectoral case studies collectively underscore that SHRM plays a transformative role in equipping Indonesian MSMEs and startups for digital resilience. Rather than serving as a mere administrative function, HRM when approached strategically becomes a core enabler of innovation, adaptability, and sustainable competitive advantage in the evolving digital economy.

Overall, SHRM represents a dynamic and evolving discipline that plays a critical role in driving organizational success. By integrating HR strategies with business objectives, organizations can enhance employee performance, foster innovation, and achieve long-term sustainability in an increasingly competitive environment.

2.2. Relevant theory

The Dynamic Capabilities Theory (Teece et al., 1997) explains how organizations can sustain competitive advantage by adapting to environmental changes, particularly technological advancements. This theory suggests that firms must develop the ability to sense opportunities, seize them, and transform their internal processes to remain competitive. In the context of Human Resource Management (HRM), dynamic capabilities enable companies to align their workforce strategies with rapid digital transformations. Organizations that can swiftly adopt digital HR technologies, restructure work processes, and enhance employee competencies through continuous learning will be better positioned to thrive in the digital economy.

The Resource-Based View (RBV) Theory of Barney (1991) emphasizes that an organization's sustained competitive advantage is derived from its unique resources, particularly human capital. According to this theory, employees with specialized skills, knowledge, and expertise serve as valuable, rare, inimitable, and non-substitutable (VRIN) assets that drive organizational success. In the digital era, RBV underscores the necessity of investing in digital training, talent development, and strategic workforce management to enhance productivity and long-term performance. Companies that prioritize human capital development through digital learning platforms and AI-driven HR practices can strengthen their market position and foster innovation.

The Technology and Human Resource Management Theory of Strohmeier (2014) highlights the evolving role of technology in HRM, focusing on how digitalization transforms workforce management. This theory explains the integration of HR analytics, artificial intelligence, and automation in various HR functions, such as recruitment, training, performance management, and employee engagement. By leveraging advanced technologies, organizations can improve decision-making, enhance operational efficiency, and optimize talent management processes. The increasing reliance on HR analytics allows firms to predict workforce trends, assess employee performance more accurately, and design personalized development programs, ultimately driving workforce productivity and organizational effectiveness.

2.3. Previous research

Previous research on Strategic Human Resource Management (SHRM) in the era of the digital economy highlights both challenges and opportunities for local enterprises. Raman and Khan (2024) argue that SHRM must evolve to address the complexities of digitalization and economic fluctuations, particularly for businesses operating in local markets. The digital transformation of SMEs presents both hurdles and potential growth avenues, as discussed by Faeni and Wibisana (2024), who emphasize the need for human resource development to support digital adaptation. Additionally, Zhang and Chen (2023) highlight how digital HRM transformation can improve operational efficiency and workforce agility, which is crucial for local enterprises striving to remain competitive.

One of the major challenges in implementing SHRM within the digital economy is ensuring that local enterprises can adapt to technological advancements while maintaining workforce stability. Cahyadi et al. (2022) find that leadership styles and high-involvement HRM practices significantly influence employee performance in SMEs, demonstrating that effective HR strategies can help local businesses navigate the digital shift. Similarly, Gileva et al. (2021) introduce the concept of digital maturity, arguing that HRM must facilitate continuous upskilling and reskilling to enhance business resilience in the digital economy. Furthermore, Adiazmil et al. (2024) stress the importance of strategic HR planning in supporting digital transformation, ensuring that local enterprises can optimize human capital while integrating new technologies.

Despite these challenges, digitalization also presents significant opportunities for local enterprises. Minghui and Chelliah (2024) show that strategic management, when combined with digital HR practices, can enhance organizational performance, positioning businesses for long-term success. Ferdousi and Abedin (2023) further explore how SHRM contributes to shared value creation, emphasizing that human capital strategies aligned with digital advancements can drive innovation and business sustainability. Additionally, Donnelly and Johns (2021) propose a framework for remote work integration, which is particularly beneficial for local enterprises aiming to expand their talent pool beyond geographical constraints.

In response to the Society 5.0 paradigm, Misbah and Budiyanto (2025) suggest that local enterprises must integrate SHRM with technological innovations to maintain a competitive edge. Meanwhile, Alsaadat (2018) highlights the impact of HR technology on training and workforce development, demonstrating that local businesses can leverage digital tools to enhance employee capabilities. Lastly, Farliana et al. (2023) discuss Indonesia's digital economic landscape, emphasizing that HRM strategies must be tailored to regional challenges and opportunities to support sustainable business growth.

2.4. Conceptual framework & hypothesis

As The Figure 1 show, the adoption of digital technology in HRM improves workforce productivity by automating administrative processes, reducing manual workloads, and enhancing communication. Digital HR tools, such as cloud-based HR systems and AI-driven performance tracking, streamline operations and allow employees to focus on strategic tasks. These advancements contribute to efficiency, minimize errors, and create a more productive work environment.

H1: Digital Technology Adoption in HRM (X_1) has a positive impact on Workforce Productivity (Y_1).

Digital transformation in HRM enhances overall organizational performance by improving talent management, decision-making, and operational agility. The integration of technology facilitates better workforce planning, optimizes HR functions, and strengthens employee engagement, ultimately leading to improved business outcomes. Organizations that effectively utilize digital HRM systems gain a competitive advantage by adapting more quickly to industry demands. Digital adoption in HRM plays a crucial role in enhancing workforce productivity by streamlining HR processes, improving data-driven decision-making, and fostering employee engagement. According to Suwaji et al. (2024), the implementation of HR technology significantly transforms human resource management by automating administrative tasks, enhancing recruitment processes, and improving talent retention strategies, ultimately leading to increased organizational efficiency and workforce productivity. Similarly, Salam and Salam and

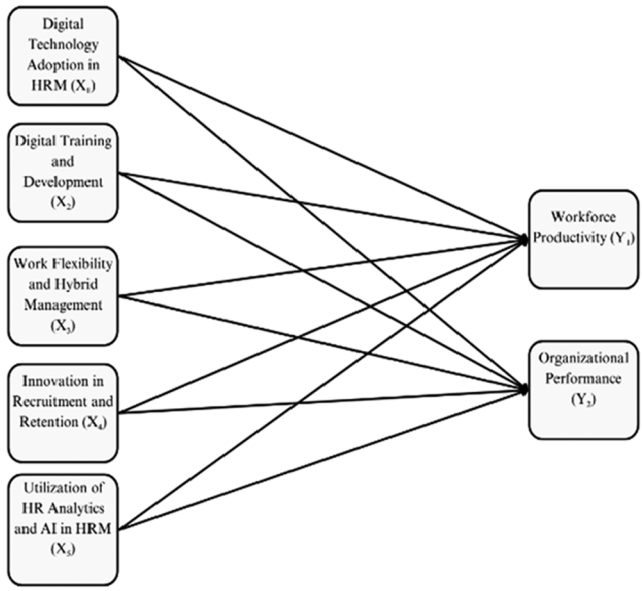

Figure 1. Conceptual framework.

Munawir (2024) emphasize that the integration of digital tools in HRM enables organizations to optimize operational efficiency, reduce workload redundancies, and enhance employee performance through real-time analytics and strategic workforce planning. These findings underscore the positive impact of digital transformation in HRM on workforce productivity, reinforcing its role in driving organizational success in the digital era.

H2: Digital Technology Adoption in HRM (X_1) has a positive impact on Organizational Performance (Y_2).

Digital training and development initiatives enable employees to acquire new skills and improve job performance through accessible and flexible learning platforms. Online training modules, virtual simulations, and AI-driven learning systems support continuous professional growth, ensuring that employees stay competent and adaptable. This fosters higher productivity as employees can efficiently apply newly acquired knowledge to their roles. The adoption of digital technology in HRM significantly enhances organizational performance by fostering employee development, optimizing HR functions, and improving strategic decision-making. Sharma and Ahmad (2022) highlight that organizations leveraging technology-driven HRM practices experience improved efficiency, talent management, and operational agility, leading to enhanced overall performance. Vrontis et al. (2023) further emphasize that artificial intelligence, robotics, and advanced technologies in HRM streamline recruitment, training, and workforce analytics, enabling organizations to adapt quickly to market changes and maintain a competitive edge. Additionally, Turulja and Bajgoric (2018) argue that integrating information technology with knowledge and human resource management enhances organizational learning, innovation, and productivity, ultimately driving superior performance outcomes. These findings underscore the critical role of digital HRM adoption in shaping a high-performing, future-ready organization.

H3: Digital Training and Development (X_2) has a positive impact on Workforce Productivity (Y_1).

Investing in digital training enhances organizational performance by equipping employees with the necessary expertise to navigate evolving business landscapes. A well-trained workforce contributes to higher efficiency, reduced operational disruptions, and better service delivery. Organizations that prioritize digital learning gain long-term benefits through improved workforce capabilities and sustained business growth. Digital training and development initiatives play a crucial role in enhancing workforce productivity by equipping employees with relevant skills and competencies needed in the evolving digital landscape. According to Juliadi et al. (2023), organizations that implement structured digital training programs experience increased employee efficiency, minimized errors, and improved task performance, which directly contributes to higher productivity levels. Similarly, Ferdiansah (2024)

highlights that continuous learning opportunities, such as e-learning platforms and virtual training sessions, enable employees to stay updated with industry trends, enhancing their ability to adapt and perform effectively. These studies affirm that investing in digital training not only boosts individual employee performance but also drives overall workforce productivity, ensuring long-term organizational success.

H4: Digital Training and Development (X_2) has a positive impact on Organizational Performance (Y_2).

Flexible work arrangements and hybrid management models support workforce productivity by promoting work-life balance and reducing unnecessary commuting time. Employees who have greater control over their work schedules tend to be more motivated and focused, leading to increased efficiency. Digital collaboration tools further enhance productivity by enabling seamless communication and task coordination, regardless of location. Digital training and development contribute significantly to organizational performance by fostering a skilled workforce, enhancing adaptability, and improving overall efficiency. Cera and Kusaku (2020) emphasize that continuous training and development initiatives positively influence organizational success by equipping employees with relevant competencies, promoting innovation, and strengthening corporate culture. Similarly, Sousa and Rocha (2019) highlight that digital learning initiatives, such as e-learning platforms and AI-driven training programs, enable employees to develop the necessary skills for digital transformation, thereby improving organizational agility and competitiveness. By integrating digital training into HR strategies, organizations can enhance employee performance, streamline operations, and ultimately drive sustainable business growth.

H5: Work Flexibility and Hybrid Management (X_3) has a positive impact on Workforce Productivity (Y_1).

Organizations that implement hybrid work models experience improved performance due to increased employee satisfaction, cost savings, and enhanced operational agility. By allowing a mix of remote and in-office work, companies can attract and retain top talent while maintaining productivity. Hybrid management strategies foster innovation, adaptability, and resilience, which are crucial for sustaining organizational success. Work flexibility and hybrid management models contribute to increased workforce productivity by providing employees with greater autonomy over their work schedules, reducing commuting stress, and enhancing job satisfaction. Organizations that implement hybrid work arrangements benefit from improved engagement and efficiency, as employees can balance personal and professional responsibilities more effectively (Givan, 2024). Moreover, hybrid models facilitate seamless collaboration through digital tools, ensuring that remote and on-site employees remain connected and productive. A study by Tran (2022) highlights that hybrid work arrangements enhance productivity, particularly in agile work environments, by enabling employees to work in settings that optimize focus and creativity. These findings suggest that hybrid management is a strategic approach to sustaining high levels of workforce performance while fostering innovation and adaptability.

H6: Work Flexibility and Hybrid Management (X_3) has a positive impact on Organizational Performance (Y_2).

Innovative recruitment and retention strategies help organizations secure top talent and maintain a motivated workforce. AI-driven recruitment, data-driven talent acquisition, and personalized retention programs ensure that employees are well-matched to their roles and engaged in their work. These improvements contribute to better job performance, reduced turnover, and higher workforce productivity. Work flexibility and hybrid management models significantly enhance organizational performance by fostering employee engagement, optimizing productivity, and improving talent retention. Naqshbandi et al. (2024) highlight that hybrid work arrangements promote higher work engagement and job performance by offering employees greater autonomy, leading to increased job satisfaction and organizational commitment. Similarly, White et al. (2021) emphasize that effective hybrid management practices, such as structured communication, performance-based evaluations, and digital collaboration tools, contribute to improved operational efficiency and overall business outcomes. By leveraging flexible work models and innovative management strategies, organizations can create a dynamic and resilient workforce, ultimately driving long-term success and competitive advantage.

H7: Innovation in Recruitment and Retention (X_4) has a positive impact on Workforce Productivity (Y_1).

A strong recruitment and retention strategy enhances organizational performance by ensuring workforce stability and reducing hiring costs. Organizations that implement advanced recruitment technologies and employee retention initiatives benefit from a more skilled and committed workforce. This leads to improved operational efficiency, better customer service, and a stronger market position.

Innovative recruitment and retention strategies significantly contribute to workforce productivity by attracting top talent and ensuring long-term employee commitment. Jose and Asha (2019) highlight that leveraging advanced recruitment technologies, such as AI-driven candidate screening and predictive analytics, helps organizations secure highly skilled employees who align with business needs, ultimately boosting productivity. Similarly, Mohlala et al. (2024) emphasize that the integration of HR information systems (HRIS) and AI-driven talent management enhances workforce efficiency by streamlining hiring processes, improving employee engagement, and reducing turnover rates. These findings underscore that organizations investing in strategic recruitment and retention innovations can achieve higher workforce productivity, operational efficiency, and sustained competitive advantage.

H8: Innovation in Recruitment and Retention (X_4) has a positive impact on Organizational Performance (Y_2).

The use of HR analytics and AI in HRM enhances workforce productivity by enabling data-driven decision-making, optimizing performance management, and improving employee experiences. AI-powered insights help HR teams identify skill gaps, predict workforce trends, and personalize employee development programs. These capabilities lead to a more efficient and productive workforce. Innovation in recruitment and retention plays a crucial role in enhancing organizational performance by ensuring workforce stability, reducing turnover, and improving employee engagement. Gyurák Babeľová et al. (2020) highlight that organizations implementing sustainable HRM strategies tailored to different generational groups experience improved retention rates and overall performance. Igbinoba et al. (2022) further emphasize that data-driven retention strategies contribute to a more engaged and productive workforce, leading to enhanced organizational outcomes. Additionally, Babel'ová et al. (2020) argue that advanced recruitment and retention approaches, including HR analytics and AI-driven decision-making, allow organizations to align talent management with business objectives, fostering long-term growth and competitive advantage. These findings underscore the significance of innovative recruitment and retention strategies in driving organizational success and workforce optimization.

H9: Utilization of HR Analytics and AI in HRM (X_5) has a positive impact on Workforce Productivity (Y_1).

HR analytics and AI contribute to improved organizational performance by streamlining HR processes, enhancing workforce planning, and supporting strategic decision-making. By leveraging data analytics, organizations can optimize talent management, improve employee engagement, and reduce inefficiencies. These advancements create a more agile and competitive organization capable of sustaining long-term success. The utilization of HR analytics and AI in HRM significantly enhances workforce productivity by enabling data-driven decision-making, optimizing HR functions, and improving employee experiences. Tshabalala (2023) highlights that AI-powered tools streamline HR operations, automate repetitive tasks, and provide predictive insights that help organizations manage talent more effectively. Veshne and Jamnani (2024) further emphasize that AI-driven workforce analytics improve employee performance tracking, enhance engagement strategies, and personalize training programs, ultimately leading to a more efficient and productive workforce. Similarly, Nawaz et al. (2024) argue that the integration of AI in HRM fosters proactive workforce planning, minimizes inefficiencies, and enhances overall employee performance. These findings affirm that leveraging HR analytics and AI not only optimizes workforce productivity but also strengthens an organization's ability to remain agile and competitive in a rapidly evolving business environment.

H10: Utilization of HR Analytics and AI in HRM (X_5) has a positive impact on Organizational Performance (Y_2).

HR analytics and AI contribute to improved organizational performance by streamlining HR processes, enhancing workforce planning, and supporting strategic decision-making. By leveraging data analytics, organizations can optimize talent management, improve employee engagement, and reduce inefficiencies. These advancements create a more agile and competitive organization capable of sustaining

long-term success. The utilization of HR analytics and AI in HRM significantly enhances organizational performance by optimizing workforce management, improving decision-making, and driving strategic growth. Zehir et al. (2020) highlight that AI-powered HR analytics enable organizations to process large volumes of employee data, providing insights that improve talent acquisition, retention, and performance management. Similarly, Huang et al. (2023) emphasize that personalized HRM through AI enhances employee engagement, aligns workforce capabilities with business goals, and fosters a culture of continuous improvement. Arora et al. (2021) further argue that AI-driven HR analytics streamline HR operations, reduce administrative burdens, and support data-driven strategic planning, ultimately contributing to higher organizational efficiency and competitiveness. These findings underscore that leveraging AI and HR analytics allows organizations to build a more agile, responsive, and high-performing workforce, driving long-term business success.

While the hypotheses posited in this study predominantly assume positive relationships between digital HRM practices and organizational outcomes, it is essential to recognize that such assumptions may inadvertently oversimplify the complex dynamics inherent in MSMEs and startups within Indonesia's digital economy. The effectiveness of digital HRM initiatives, such as digital training, talent acquisition platforms, or performance management systems, is likely contingent on multiple contextual variables including organizational size, resource availability, sector-specific characteristics, and the level of digital maturity. For instance, smaller MSMEs may face technological constraints or lack skilled personnel to fully leverage digital HRM tools, potentially resulting in neutral or even adverse effects on employee engagement and organizational performance. Moreover, the rapidly evolving digital landscape can introduce challenges related to change management, employee resistance, or cybersecurity risks, which might attenuate or reverse the expected positive impacts. Therefore, adopting a more nuanced and conditional approach to hypothesized relationships not merely assuming uniform positivity would enrich the empirical analysis, improve theoretical robustness, and offer more practical insights for MSMEs and startups navigating digital transformation in Indonesia.

3. Research methodology

3.1. Research design

The research employed a quantitative approach, utilizing a survey method to collect data from respondents. A structured questionnaire was designed to gather relevant information on the digital transformation of HRM practices in MSMEs and startups, specifically focusing on their impact on organizational performance and workforce productivity. The survey questions were carefully crafted to ensure alignment with the study's objectives, covering variables such as digital technology adoption in HRM, digital training and development, work flexibility, innovation in recruitment and retention, and the utilization of HR analytics and AI. The data collection process relied on online platforms to ensure accessibility and convenience for respondents. By using Google Forms and distributing the survey through widely-used media channels like WhatsApp, email, and business forums, the study aimed to reach a diverse sample of MSME and startup representatives across different regions of Indonesia. The questionnaire ensured that the responses were confidential and anonymous, reinforcing the ethical standards of the research. Additionally, the use of a cross-sectional survey design allowed the study to capture a snapshot of current HRM practices in the target organizations, providing valuable insights into the relationship between digital HR strategies and business performance.

3.2. Population and sample

This study targeted the entire population of Micro, Small, and Medium Enterprises (MSMEs) and startups in Indonesia, which amounts to 65,502,562 business entities. Of this total, approximately 65.5 million were MSMEs, and 2,562 were registered startups. Due to the large size of the population, the Slovin formula was applied to calculate an appropriate sample size, resulting in 400 respondents. The participants were purposively selected based on specific inclusion criteria, such as business scale, industry sector, and operational status, to ensure a broad representation across various business

Table 1. Respondent demographics.

Variables	Frequency	Percentage (%)
Business type		
MSMEs	300	75%
Startups	100	25%
Business age		
1–3 years	180	45%
4–7 years	140	35%
8+ years	80	20%
Business scale		
Micro	200	50%
Small	100	25%
Medium	50	12.5%
Startup	100	25%
Industry sector		
Retail & Trade	120	30%
Manufacturing	80	20%
Services	100	25%
Technology	70	17.5%
Others	30	7.5%
Digital adoption		
Yes	350	87.5%
No	50	12.5%

categories. As detailed in Table 1, the sample included a balanced distribution of MSMEs and startups, as well as variation in business age, industry sector, and levels of digital adoption. To ensure the sample was representative of the population, several steps were taken during the selection process. First, the sample was designed to reflect the proportion of MSMEs and startups within the total population, with efforts made to include both business types in the final sample. Moreover, the researchers paid close attention to business characteristics, ensuring a diverse representation in terms of industry sectors, business sizes (micro, small, and medium), and the digital adoption levels of the businesses. This was intended to provide a comprehensive view of the diverse entrepreneurial landscape in Indonesia. Additionally, demographic variables such as business age and operational status were incorporated into the sample selection criteria to enhance the diversity and relevance of the data collected. However, certain limitations should be acknowledged. The purposive selection method, combined with the use of online data collection, may have led to an underrepresentation of businesses located in rural or remote areas, particularly those with limited internet access or lower digital literacy. Furthermore, the relatively small proportion of startups in the national population compared to their sample representation (25%) could introduce a degree of sampling bias. While efforts were made to ensure the sample captured the heterogeneity of the population, the methodology may still have limitations in fully representing all business categories across Indonesia. All participants provided informed consent before taking part in the study. Anonymity and confidentiality were strictly maintained throughout the research process. The study utilized a quantitative research method with a survey design, which was deemed suitable for this type of research. The survey was distributed online to managers of MSMEs and startups in Indonesia. While the use of the Slovin formula to determine the sample size was noted, the authors did not provide specific details on how participants were selected or the steps taken to ensure sample representativeness. This absence of detailed explanation is a potential weakness of the study, especially considering that online distribution may lead to sampling bias.

The justification for the use of Slovin's formula lies in its applicability for estimating sample size when the population size is large and the population variance is unknown. This method was selected to achieve a manageable and statistically significant sample size given the immense number of MSMEs and startups in Indonesia. A 5% margin of error was deemed sufficient to capture key trends in digital HRM practices without excessive sampling demands. However, this approach assumes random sampling, which contrasts with the purposive sampling applied in the study an inherent methodological limitation that must be acknowledged. Furthermore, this research utilized self-reported data collected via structured online questionnaires, which raises the potential for self-reporting bias. Respondents may have provided socially desirable answers or misreported information related to their digital maturity or HRM practices. Another potential limitation involves common method variance (CMV), which can occur when data for

both independent and dependent variables are gathered from the same source using the same method. To mitigate CMV, procedural remedies were applied, such as ensuring respondent anonymity and varying item formats. However, no statistical test such as Harman's single factor test was conducted, which future research should consider to strengthen the robustness of findings.

Additionally, the dominance of the retail and trade (30%) and manufacturing (20%) sectors in the sample reflects their greater visibility and accessibility in digital transformation initiatives, such as e-commerce integration and production automation. These sectors tend to respond more actively to digital adoption efforts, making them more likely to be included in research focused on strategic human resource management in the digital era. However, this concentration affects the generalizability of the findings, as businesses in other sectors such as technology, services, or creative industries experience distinct challenges, operational models, and HRM practices. As a result, the study's insights are more representative of traditional or semi-digitized enterprises, rather than digitally native or service-based businesses. To address this limitation, future research should adopt stratified random sampling or apply weighting adjustments to ensure a more proportionate representation across industry sectors, thereby enhancing the external validity and applicability of the findings.

3.3. Research instruments

Research data were obtained from questionnaires distributed to respondents, with participants' responses serving as the primary data source for this study. The questionnaire was designed using a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). The research instrument consisted of 36 question items covering six research variables, with each variable measured by 6 items. The independent variables include Digital Technology Adoption in HRM (X_1), Digital Training and Development (X_2), Work Flexibility and Hybrid Management (X_3), Innovation in Recruitment and Retention (X_4), and Utilization of HR Analytics and AI in HRM (X_5), while the dependent variables consist of Workforce Productivity (Y_1) and Organizational Performance (Y_2). Each measurement item was adapted from validated sources to ensure the reliability and validity of the instrument in assessing the impact of digital HRM strategies on business performance.

Research data were obtained from questionnaires distributed to respondents, with participants' responses serving as the primary data source for this study. The questionnaire was designed using a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). The research instrument consisted of 42 question items covering six research variables, with each variable measured by six items. The research instrument is described briefly. It includes 42 items measured on a 5-point Likert scale, and the items are adapted from existing studies. Importantly, the items were translated and culturally adapted for the Indonesian context to ensure clarity and relevance for local respondents. The independent variables include Digital Technology Adoption in HRM (X_1), which comprises Infrastructure Readiness (availability of digital tools and system integration) (Bondarouk & Brewster, 2016), Employee Digital Competency (familiarity with HR digital systems and training) (Noe et al., 2014), and Organizational Support for Digitalization (management commitment and investment in HR technology) (Tambe et al., 2019). Digital Training and Development (X_2) includes Training Accessibility (availability of online training and employee participation) (Aguinis & Kraiger, 2009), Skill Enhancement (digital skill improvement and application of acquired skills) (Cascio & Montealegre, 2016), and Training Effectiveness (impact on job performance and employee satisfaction) (Marler & Boudreau, 2017). Work Flexibility and Hybrid Management (X_3) consists of Flexible Work Arrangements (remote work options and flexible schedules) (Kossek & Thompson, 2016), Team Collaboration (use of digital communication tools and employee engagement) (Gilson et al., 2015), and Work-Life Balance (reduced stress and perception of balance) (Greenhaus & Allen, 2011). For recruitment and analytics, Innovation in Recruitment and Retention (X_4) consists of Digital Recruitment Tools (AI in candidate selection and effectiveness of job postings) (Lievens & Chapman, 2019), Employee Engagement in Retention (career development opportunities and feedback systems) (Albrecht et al., 2018), and Employer Branding (online reputation and social media utilization for talent acquisition (Theurer et al., 2018). Utilization of HR Analytics and AI in HRM (X_5) includes HR Data-Driven Decision-Making (analytics in workforce planning and performance evaluation) (Marler & Boudreau, 2017), AI in HR Operations (automation in HR processes and predictive analytics) (Tambe et al., 2019), and

Ethical AI Implementation (transparency in AI-based HR decisions and employee trust) (Berkelaar, 2017). The dependent variables include Workforce Productivity (Y_1), measured by Task Efficiency (task completion speed and error reduction) (Fournier et al., 2024), Employee Performance (goal achievement and contributions to team objectives) (Tarique & Schuler, 2010), and Job Satisfaction and Motivation (engagement levels and enthusiasm for work) (Lemonaki et al., 2021). Organizational Performance (Y_2) includes Operational Efficiency (cost reduction through digital HRM and increased process efficiency) (Vardarlier, 2020), Financial Performance (revenue growth and returns on HR technology investments) (Sparrow & Makram, 2015), and Innovation and Adaptability (responsiveness to market changes and adoption of new HR strategies) (Jiang & Messersmith, 2018). Each measurement item was adapted from validated sources to ensure the reliability and validity of the instrument in assessing the impact of digital HRM strategies on business performance.

3.4. Data analysis

This research employs a confirmatory approach using a quantitative method with Structural Equation Modeling (SEM) via Smart PLS to analyze and validate the proposed model, as SEM allows for the simultaneous evaluation of relationships between latent constructs. The study involves three types of data analysis: descriptive analysis, measurement model analysis, and hypothesis testing. The descriptive analysis includes examining the mean values, standard deviations, and factor loadings of the variables to provide an overview of the data distribution and ensure the adequacy of the variables in representing the constructs. The key variables in this study are five independent variables Digital Technology Adoption in HRM (X_1), Digital Training and Development (X_2), Work Flexibility and Hybrid Management (X_3), Innovation in Recruitment and Retention (X_4), and Utilization of HR Analytics and AI in HRM (X_5) and two dependent variables Workforce Productivity (Y_1) and Organizational Performance (Y_2). The measurement model analysis evaluates the validity and reliability of the measurement items used for these constructs, ensuring that they are accurately measured by their respective indicators. Finally, hypothesis testing using Smart PLS is conducted to assess the significance of the relationships between the independent and dependent variables, providing empirical insights into the impact of digital HRM strategies on workforce productivity and organizational performance. This approach ensures a comprehensive evaluation of the proposed model and its ability to explain the relationships between digital HRM strategies and business outcomes.

3.5. Ethics statement

This study has been approved by the Ethics Committee of Palangka Raya University. Prior to data collection, all respondents were given detailed information about the purpose, procedures, and their rights as research participants. Written informed consent was obtained from each participant prior to data collection. Participants were guaranteed the right to withdraw from the study at any time without any consequences. All data was stored securely, with limited access to the research team, to ensure confidentiality and privacy throughout the research process.

4. Results and analysis

4.1. Descriptive analysis

The descriptive analysis in this study provides an overview of respondents' perceptions regarding various constructs related to digital transformation in human resource management. Using a five-point Likert scale, where 1 represents 'strongly disagree' and 5 represents 'strongly agree,' the analysis includes the mean, median, standard deviation, and loading factor for each item. The mean values indicate that respondents generally agree with the statements regarding digital transformation aspects, such as digital technology adoption, digital training, work flexibility, and HR analytics. The standard deviation reflects some variation in responses, suggesting a mix of strong and moderate agreements among respondents. The loading factors demonstrate the strength of the association between each item and its respective

construct, confirming the reliability and validity of the measurements used in this study. This analysis is essential to assess the central tendency, variability, and overall perception of respondents toward the constructs of interest in the context of HRM digital transformation (Kline, 2023). The descriptive analysis of this study can be seen in Table 2, below.

For Digital Technology Adoption in HRM (X_1), the mean scores range between 3.710 and 3.935, with a consistent median of 4.000, indicating a general agreement among respondents. The standard deviation varies from 1.111 to 1.156, reflecting a moderate spread in responses. The loading factor values, ranging from 0.739 to 0.776, suggest that each item contributes significantly to the construct. In the case of Digital Training and Development (X_2), the mean values fall between 3.578 and 3.680, showing a slightly lower agreement compared to digital technology adoption, but still indicating a positive perception. The median remains steady at 4.000, while the standard deviation ranges from 1.133 to 1.197, signifying moderate dispersion. The loading factor values, varying between 0.730 and 0.806, confirm that all items hold substantial significance in defining the construct. The construct of Work Flexibility and Hybrid Management (X_3) records mean values between 3.748 and 3.973, with a median consistently at 4.000, showing that respondents perceive work flexibility positively. The standard deviation ranges from 0.992 to 1.093, and the loading factors vary between 0.722 and 0.822, demonstrating that all items are relevant in representing the construct. Meanwhile, Innovation in Recruitment and Retention (X_4) exhibits mean values between 3.482 and 3.853, reflecting slightly lower agreement levels. The median remains 4.000, while the standard deviation fluctuates between 1.094 and 1.212. The loading factor values, spanning from 0.711 to 0.803, indicate that the construct is well-supported. For Utilization of HR Analytics and AI in HRM (X_5), the mean scores range from 3.375 to 3.495, with a median mostly at 4.000, except

Table 2. Descriptive analysis.

	Item	Mean	Median	STD	Loading factor
Digital technology adoption in HRM (X_1)	X1.1	3.935	4.000	1.112	0.739
	X1.2	3.785	4.000	1.153	0.776
	X1.3	3.783	4.000	1.156	0.751
	X1.4	3.728	4.000	1.137	0.742
	X1.5	3.710	4.000	1.134	0.764
	X1.6	3.712	4.000	1.111	0.743
Digital training and development (X_2)	X2.1	3.672	4.000	1.151	0.761
	X2.2	3.615	4.000	1.161	0.775
	X2.3	3.615	4.000	1.197	0.801
	X2.4	3.680	4.000	1.167	0.730
	X2.5	3.678	4.000	1.178	0.806
	X2.6	3.578	4.000	1.133	0.793
Work flexibility and hybrid management (X_3)	X3.1	3.973	4.000	1.052	0.822
	X3.2	3.865	4.000	1.066	0.791
	X3.3	3.748	4.000	1.093	0.777
	X3.4	3.822	4.000	1.042	0.722
	X3.5	3.908	4.000	1.071	0.726
	X3.6	3.925	4.000	0.992	0.738
Innovation in recruitment and retention (X_4)	X4.1	3.708	4.000	1.094	0.711
	X4.2	3.675	4.000	1.149	0.764
	X4.3	3.610	4.000	1.212	0.802
	X4.4	3.482	4.000	1.192	0.803
	X4.5	3.853	4.000	1.181	0.724
	X4.6	3.542	4.000	1.163	0.799
Utilization of HR analytics and AI in HRM (X_5)	X5.1	3.495	4.000	1.239	0.817
	X5.2	3.375	3.000	1.220	0.809
	X5.3	3.385	3.000	1.217	0.796
	X5.4	3.493	4.000	1.206	0.834
	X5.5	3.415	3.000	1.299	0.839
	X5.6	3.480	4.000	1.183	0.823
Workforce productivity (Y_1)	Y1.1	3.237	3.000	1.359	0.845
	Y1.2	3.188	3.000	1.355	0.844
	Y1.3	3.232	3.000	1.354	0.848
	Y1.4	3.100	3.000	1.387	0.798
	Y1.5	3.200	3.000	1.380	0.826
	Y1.6	3.337	3.000	1.352	0.803
Organizational Performance (Y_2)	Y2.1	2.958	3.000	1.121	0.761
	Y2.2	2.982	3.000	1.186	0.827
	Y2.3	2.953	3.000	1.138	0.768
	Y2.4	2.962	3.000	1.156	0.827
	Y2.5	2.830	3.000	1.121	0.762
	Y2.6	2.987	3.000	1.180	0.768

for a few items at 3.000, suggesting mixed responses. The standard deviation varies from 1.183 to 1.299, indicating greater variability in perceptions. The loading factor values, ranging from 0.796 to 0.839, confirm the construct's reliability. Regarding Workforce Productivity (Y_1), the mean scores fall between 3.100 and 3.337, with a median of 3.000, suggesting moderate perceptions. The standard deviation fluctuates between 1.352 and 1.387, while the loading factors, between 0.798 and 0.848, highlight the construct's significance. Lastly, for Organizational Performance (Y_2), the mean values range from 2.830 to 2.987, with a median of 3.000, reflecting the lowest perception levels among the constructs. The standard deviation varies from 1.121 to 1.186, while the loading factor values, between 0.761 and 0.827, still indicate strong construct validity.

4.2. Measurement model

The measurement model represents the relationship between latent constructs and their observed indicators, serving as a foundational component in structural equation modeling (SEM). In this study, the measurement model is visualized to depict how each latent variable such as digital leadership, employee engagement, and organizational performance is measured through multiple reflective indicators. Each indicator is connected to its respective construct through factor loadings, which illustrate the strength of the relationship between the observed variable and the underlying latent factor. This model facilitates (Figure 2) an initial validation of the theoretical framework by ensuring that the constructs are accurately and reliably measured. Once the measurement model is specified and illustrated, its evaluation is conducted using several key metrics to confirm its reliability and validity.

In evaluating the measurement model, several key metrics are employed to ensure its reliability and validity. Cronbach's alpha is used to assess the internal consistency of the constructs, indicating how well the items within a construct correlate with each other. According to Nunnally and Bernstein (1994), a Cronbach's alpha value above 0.7 is considered acceptable for demonstrating reliability. Composite reliability (rho_a) measures the overall reliability of the construct, accounting for factor loadings and error variances, and values greater than 0.7 are generally considered adequate (Fornell & Larcker, 1981). Similarly, composite reliability (rho_c) provides a more accurate reflection of reliability and ensures the construct's internal consistency, where values above 0.7 are deemed satisfactory (Sarstedt et al., 2022). Lastly, Average Variance Extracted (AVE) is utilized to assess the convergent validity of the construct, indicating the proportion of variance in the indicators explained by the construct itself. Fornell and Larcker (1981) suggest that an AVE value greater than 0.5 is desirable, meaning that the construct explains more than half of the variance in its indicators. Together, these metrics provide a comprehensive evaluation of the measurement model's robustness and consistency, ensuring both reliability and validity in the data.

The measurement model assessment provides insights into the reliability and validity of the constructs used in this study (see Table 3 above). The Cronbach's alpha values for all constructs are above 0.7, indicating strong internal consistency. Specifically, the Digital Technology Adoption in HRM (X1) has a Cronbach's alpha of 0.847, Digital Training and Development (X2) shows a value of 0.870, and Innovation in Recruitment and Retention (X4) has a Cronbach's alpha of 0.861, all suggesting acceptable reliability. Furthermore, the Utilization of HR Analytics and AI in HRM (X5) demonstrates a high Cronbach's alpha value of 0.903, indicating excellent internal consistency. The Composite Reliability (rho_a) and Composite Reliability (rho_c) values also reinforce these findings, with all constructs exceeding the threshold of 0.7, which is considered acceptable (Hair et al., 2010). For example, the Workforce Productivity (Y1) construct shows a rho_a of 0.908 and a rho_c of 0.929, indicating excellent composite reliability. Additionally, the Average Variance Extracted (AVE) values are all above the recommended threshold of 0.5, indicating that the constructs effectively capture the variance in their indicators. For instance, Workforce Productivity (Y1) has an AVE of 0.685, while Utilization of HR Analytics and AI in HRM (X5) has an AVE of 0.672, both demonstrating strong convergent validity. These results confirm the robustness of the measurement model in terms of reliability and validity, ensuring that the constructs used in the study are appropriate and reliable for further analysis.

To further assess the construct validity of the measurement model, it is essential to evaluate discriminant validity, which confirms that each construct represents a conceptually distinct domain. Discriminant validity ensures that a latent variable is not only empirically different from other variables but also

measures what it is intended to measure without overlapping conceptually similar constructs. This step is particularly important in studies involving multiple interrelated constructs, as it safeguards the integrity of the theoretical model. In this study, discriminant validity was examined using two widely accepted approaches: the Fornell-Larcker criterion (Fornell & Larcker, 1981) and the Heterotrait-Monotrait Ratio of Correlations (HTMT), as proposed by Henseler et al. (2015).

The measurement model for discriminant validity is evaluated using two methods: the Heterotrait-Monotrait (HTMT) ratio and the Fornell-Larcker criterion, like Table 4 above. The HTMT ratio is used to assess the degree of correlation between constructs, where a value greater than 0.85 suggests potential issues with discriminant validity. In this study, the HTMT values for all pairs of constructs remain below 0.85, indicating adequate discriminant validity between the variables. For example, the HTMT value between Digital Technology Adoption in HRM (X1) and Workforce Productivity (Y1) is 0.352, and

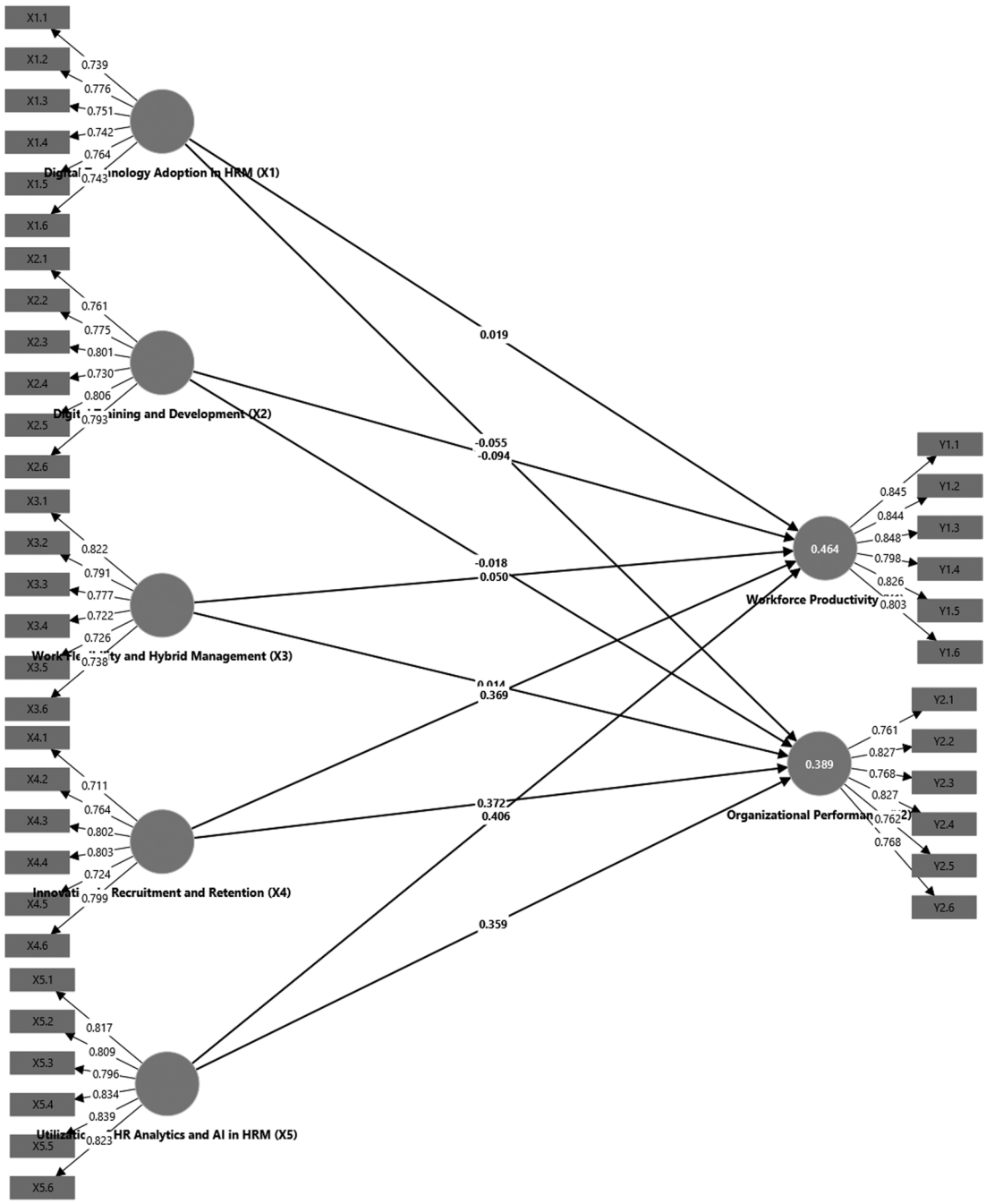

Figure 2. Measurement Model.

between Digital Training and Development (X2) and Work Flexibility and Hybrid Management (X3), it is 0.682, both of which are below the threshold, confirming that the constructs are distinct from one another. Additionally, the Fornell-Larcker criterion is used to further verify discriminant validity by comparing the square root of the Average Variance Extracted (AVE) of each construct with the correlation values between constructs. If the square root of AVE for each construct is greater than the correlation between that construct and others, discriminant validity is considered established. In this model, the diagonal values representing the square roots of AVE, such as 0.752 for Digital Technology Adoption in HRM (X1), 0.778 for Digital Training and Development (X2), and 0.827 for Workforce Productivity (Y1), are all greater than the off-diagonal correlation values, confirming that the constructs meet the Fornell-Larcker

Table 3. Measurement model construct reliability and validity.

	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
DTA (X1)	0.847	0.851	0.887	0.566
DTD (X2)	0.870	0.879	0.902	0.606
IRR (X4)	0.861	0.865	0.896	0.590
OP (Y2)	0.876	0.878	0.907	0.618
UHRAI (X5)	0.903	0.904	0.925	0.672
WFHM (X3)	0.857	0.863	0.893	0.583
WP (Y1)	0.908	0.908	0.929	0.685

Table 4. Measurement model discriminant validity.

Heterotrait-Monotrait (HTMT)							
	DTA (X1)	DTD (X2)	IRR (X4)	OP (Y2)	UHRAI (X5)	WFHM (X3)	WP (Y1)
DTA (X1)							
DTD (X2)	0.773						
IRR (X4)	0.560	0.464					
OP (Y2)	0.289	0.310	0.632				
UHRAI (X5)	0.433	0.534	0.655	0.618			
WFHM (X3)	0.645	0.682	0.567	0.357	0.502		
WP (Y1)	0.352	0.319	0.671	0.770	0.664	0.407	
Fornell Larcker criterion							
	DTA (X1)	DTD (X2)	IRR (X4)	OP (Y2)	UHRAI (X5)	WFHM (X3)	WP (Y1)
DTA (X1)	0.752						
DTD (X2)	0.659	0.778					
IRR (X4)	0.478	0.403	0.768				
OP (Y2)	0.252	0.274	0.553	0.786			
UHRAI (X5)	0.372	0.475	0.579	0.551	0.820		
WFHM (X3)	0.542	0.585	0.488	0.314	0.443	0.764	
WP (Y1)	0.311	0.289	0.600	0.687	0.604	0.366	0.827

Table 5. R square, F square, model fit.

R square		
	R-square	R-square adjusted
OP (Y2)	0.389	0.381
WP (Y1)	0.464	0.457

F Square							
	DTA (X1)	DTD (X2)	IRR (X4)	OP (Y2)	UHRAI (X5)	WFHM (X3)	WP (Y1)
DTA (X1)				0.002			0.000
DTD (X2)				0.000			0.008
IRR (X4)				0.127			0.142
OP (Y2)							
UHRAI (X5)				0.124			0.181
WFHM (X3)				0.000			0.003
WP (Y1)							

Model Fit		
	Saturated model	Estimated model
SRMR	0.061	0,070
d_ULS	3363	4,431
d_G	1175	1227
Chi-square	25,63478	26,47,216
NFI	0.766	0.759

criterion for discriminant validity. This suggests that the measurement model exhibits strong discriminant validity, ensuring that the constructs measured are distinct and not overly correlated.

4.3. R Square, F square, model fit

Based on (Hair et al., 2022), the quality of the structural model can be evaluated using three key indicators: R Square (R^2), F Square (f^2), and Model Fit. The R^2 value indicates the proportion of variance in the dependent variable explained by the independent variables, where values of 0.75, 0.50, and 0.25 are categorized as substantial, moderate, and weak respectively. The f^2 value assesses the effect size of each exogenous variable on the endogenous variable, with thresholds of 0.02 (small), 0.15 (medium), and 0.35 (large). Lastly, model fit can be examined using the Standardized Root Mean Square Residual (SRMR), where a value below 0.08 suggests a good model fit. These criteria help determine the predictive relevance and explanatory power of the proposed structural model.

As shown in Table 5 above, R-square values indicate the extent to which the independent variables Digital Technology Adoption in HRM (X1), Digital Training and Development (X2), Innovation in Recruitment and Retention (X4), Utilization of HR Analytics and AI in HRM (X5), and Work Flexibility and Hybrid Management (X3) collectively explain the variance in the dependent variables, namely Workforce Productivity (Y1) and Organizational Performance (Y2). Based on the results, the R-square value for Workforce Productivity (Y1) is 0.464, suggesting that approximately 46.4% of the variance in Workforce Productivity is explained by the model. Meanwhile, the R-square for Organizational Performance (Y2) is 0.389, meaning that 38.9% of the variation in Organizational Performance is accounted for by the independent variables. According to Hair et al. (2022), these R-square values fall within the moderate category, implying that the model has acceptable explanatory power in predicting both productivity and performance outcomes in the context of digital transformation in human resource management.

Based on the F-square (f^2) analysis, which measures the effect size of each exogenous variable on the endogenous variables, the results show varying levels of contribution to the structural model. According to Hair et al. (2022), an f^2 value of 0.02 indicates a small effect, 0.15 a medium effect, and 0.35 a large effect. In this model, Innovation in Recruitment and Retention (X4) has a medium effect size on Organizational Performance (Y2) with an f^2 value of 0.127, and also shows a medium effect on Workforce Productivity (Y1) with 0.142. Similarly, Utilization of HR Analytics and AI in HRM (X5) demonstrates a medium effect on Workforce Productivity (Y1) with 0.181, and a small-to-medium effect on Organizational Performance (Y2) with 0.124. The other variables, such as Digital Technology Adoption in HRM (X1) and Work Flexibility and Hybrid Management (X3), have negligible to small effect sizes (e.g. 0.002 and 0.000 on Y2, and 0.000 and 0.003 on Y1, respectively). Digital Training and Development (X2) shows a small effect on Workforce Productivity (Y1) with a value of 0.008. These findings suggest that Innovation in Recruitment and Retention (X4) and Utilization of HR Analytics and AI (X5) are the most influential predictors in enhancing both organizational performance and productivity in the HR digital transformation context.

Based on the model fit indices reported, the structural model demonstrates an acceptable level of fit according to the criteria suggested by Hair et al. (2022). The Standardized Root Mean Square Residual (SRMR) value for the estimated model is 0.070, which is below the recommended threshold of 0.08, indicating a good fit between the hypothesized model and the observed data. Additionally, the d_ULS (Unweighted Least Squares discrepancy) and d_G (Geodesic discrepancy) values are 4.431 and 1.227 respectively for the estimated model, and while these values are not directly interpreted with strict cut-offs, smaller values generally suggest better fit, with the saturated model providing a benchmark for comparison. The Chi-square value of 2647.216 for the estimated model is higher than that of the saturated model (2563.478), which is expected in Partial Least Squares Structural Equation Modeling (PLS-SEM), as the saturated model represents a perfect fit. Lastly, the Normed Fit Index (NFI) for the estimated model is 0.759, which, while slightly below the ideal threshold of 0.90, still indicates a moderate model fit, particularly in the context of complex models. Overall, these results suggest that the estimated model is acceptable and appropriate for further interpretation of path relationships.

4.4. Hypothesis test

To assess whether the proposed hypothesis is accepted or rejected, this study uses a significance level of 5 percent (α = 0.05). The p-value is the key indicator for evaluating the significance of the relationships between the variables. If the p-value is smaller than the specified significance level, it indicates that the relationship is statistically significant, leading to the rejection of the null hypothesis and the acceptance of the alternative hypothesis. On the other hand, if the p-value is larger than 0.05, it suggests that the relationship between the variables is not statistically significant, leading to the acceptance of the null hypothesis. This method allows for the determination of the strength and relevance of the relationships within the model. As supported by Hair et al. (2021), hypothesis testing through p-values is a crucial step in validating the model and ensuring the robustness of the research findings. Similarly, (Kline, 2023) emphasizes that statistical significance at the 5 percent level provides a reliable basis for evaluating the theoretical relationships proposed in the study.

Referring to Figure 3 and Table 6, hypothesis test offers critical insights into the relationships between the constructs and their impact on organizational performance and workforce productivity. Digital Technology Adoption in HRM (X1) does not demonstrate a significant effect on either Workforce Productivity (Y1) or Organizational Performance (Y2), with p-values of 0.709 and 0.320, respectively, both exceeding the 0.05 significance threshold. This indicates that the adoption of digital technology in HRM does not have a meaningful impact on these two outcomes within the context of this study. Similarly, Digital Training and Development (X2) does not significantly influence Organizational Performance (Y2), with a p-value of 0.742, and also has no significant effect on Workforce Productivity (Y1), with a p-value of 0.056, which remains marginally above the threshold. Work Flexibility and Hybrid Management (X3) also show no significant effect on either Workforce Productivity (Y1) or Organizational Performance (Y2), as indicated by p-values of 0.285 and 0.797, respectively. On the other hand, Innovation in Recruitment and Retention (X4) shows highly significant positive effects on both Workforce Productivity (Y1) and Organizational Performance (Y2), with p-values of 0.000 for both paths. These results strongly suggest that innovations in recruitment and retention are key drivers of enhanced organizational performance and workforce productivity. Similarly, Utilization of HR Analytics and AI in HRM (X5) also exhibits significant positive relationships with both Workforce Productivity (Y1) and Organizational Performance (Y2), with p-values of 0.000, reinforcing the critical role of data-driven HR practices in boosting business outcomes. In summary, while digital technology adoption, digital training and development, and work flexibility did not show significant effects, innovations in recruitment and retention, as well as the use of HR analytics and AI, emerge as powerful factors driving both organizational performance and workforce productivity.

5. Discussions

The findings of this study provide significant insights into the role of digital transformation in human resource management (HRM) and its impact on organizational performance and workforce productivity within the context of Micro, Small, and Medium Enterprises (MSMEs) and startups in Indonesia. Firstly, the lack of statistical significance between Digital Technology Adoption in HRM and both Workforce Productivity and Organizational Performance aligns with earlier studies that argue digital technology alone does not guarantee improved business outcomes unless accompanied by proper implementation and integration strategies (Bughin et al., 2018). The p-values of 0.709 for Workforce Productivity and 0.320 for Organizational Performance suggest that digital technology adoption might not be sufficient by itself to impact the productivity or performance of businesses. This finding could be attributed to the fact that many MSMEs and startups might lack the infrastructure or expertise to leverage digital tools effectively, which is consistent with previous research indicating that smaller firms often face challenges in adopting and utilizing technology (Vial, 2021).

Digital Training and Development shows no significant effect on Organizational Performance ($p = 0.742$) and Workforce Productivity ($p = 0.056$). This aligns with prior research emphasizing that training programs enhance employee skills but may not immediately translate into organizational-level performance improvements (P. R. Sparrow & Makram, 2015). The lack of significance in its relationship with workforce

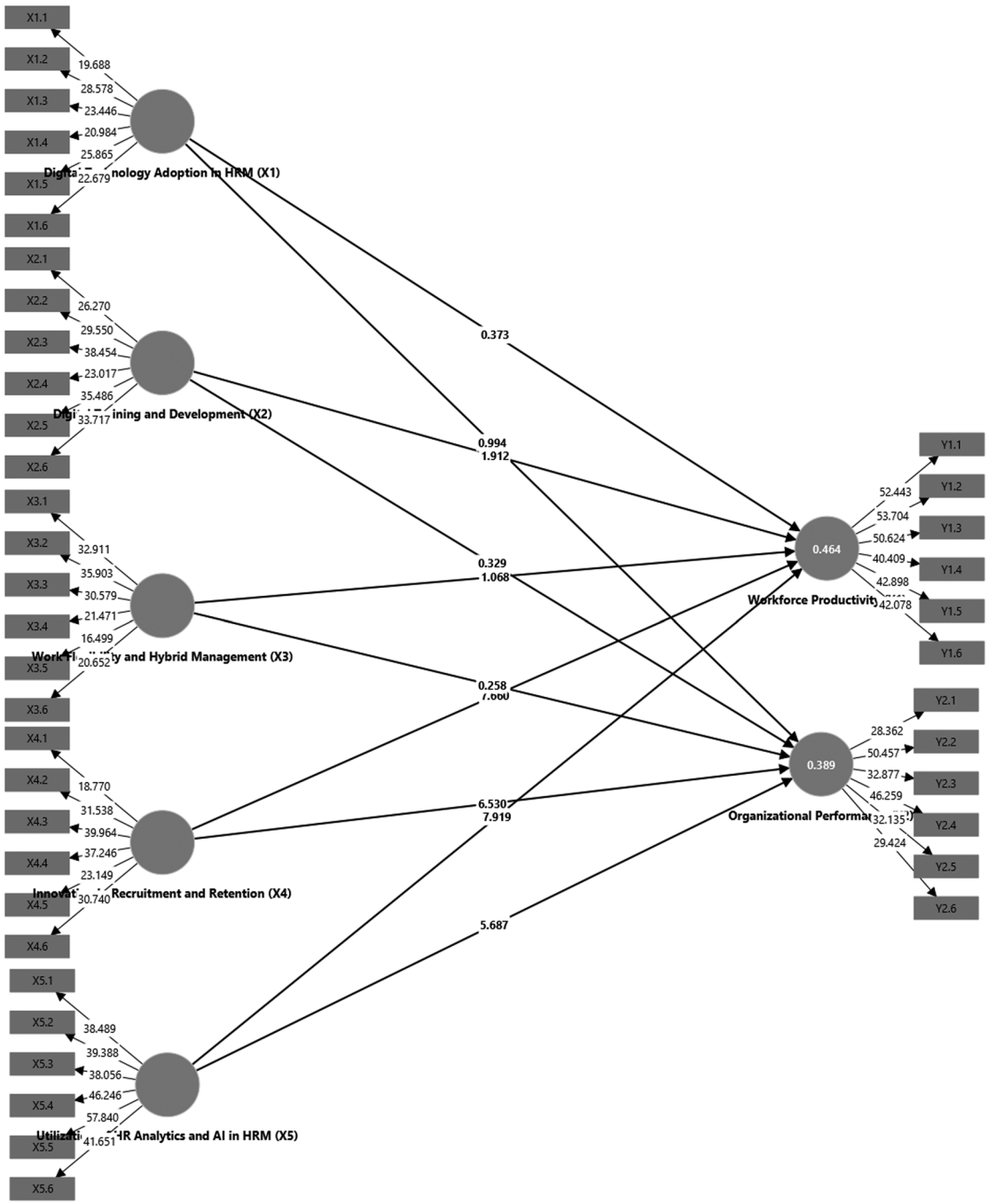

Figure 3. Bootstrapping Model (Hypothesis Test).

productivity suggests that while structured training programs may contribute to individual efficiency, a broader strategic approach is necessary to achieve a firm-wide impact. Similar findings were observed by Salas et al. (2012), who argued that although training can enhance individual performance, its direct impact on organizational performance often takes time to materialize. Furthermore Noe (2020) highlighted that the effectiveness of training programs is contingent on the alignment of training objectives with organizational goals, suggesting that training programs alone may not be sufficient without a comprehensive strategy for integrating them into the organizational culture.

Work Flexibility and Hybrid Management do not significantly impact either Workforce Productivity ($p = 0.285$) or Organizational Performance ($p = 0.797$). This result suggests that while flexible work arrangements may improve work-life balance, they do not necessarily drive measurable performance improvements. Prior research indicates that the effectiveness of hybrid work models depends on organizational

Table 6. Hypothesis test.

	Original sample	Sample mean	STD	T statistics	*p* values	Results
DTA (X1) -> WP (Y1)	0.019	0.021	0.050	0.373	0.709	Not Accepted
DTA (X1) -> OP (Y2)	−0.055	−0.052	0.055	0.994	0.320	Not Accepted
DTD (X2) -> WP (Y1)	−0.094	−0.094	0.049	1912	0.056	Not Accepted
DTD (X2) -> OP (Y2)	−0.018	−0.019	0.055	0.329	0.742	Not Accepted
WFHM (X3) -> WP (Y1)	0.050	0.050	0.047	1068	0.285	Not Accepted
WFHM (X3) -> OP (Y2)	0.014	0.013	0.054	0.258	0.797	Not Accepted
IRR (X4) -> WP (Y1)	0.369	0.369	0.048	7660	0.000	Accepted
IRR (X4) -> OP (Y2)	0.372	0.373	0.057	6530	0.000	Accepted
UHRAI (X5) -> WP (Y1)	0.406	0.407	0.051	7919	0.000	Accepted
UHRAI (X5) -> OP (Y2)	0.359	0.359	0.063	5687	0.000	Accepted

culture, managerial support, and industry-specific factors (Kossek & Thompson, 2016). The lack of significant effects in this study may be attributed to MSMEs' limited experience in managing remote and hybrid workforces effectively. Moreover, Allen et al. (2015) highlighted that hybrid work models often face challenges in fostering collaboration and maintaining team cohesion, which could potentially explain the lack of significant impact observed in this study.

The highly significant results for Innovation in Recruitment and Retention and Utilization of HR Analytics and AI are consistent with the growing body of literature suggesting that innovative HR practices and data-driven decision-making significantly enhance organizational outcomes. Both Innovation in Recruitment and Retention and Utilization of HR Analytics and AI yielded p-values of 0.000, indicating strong positive effects on both Workforce Productivity and Organizational Performance. These findings support previous research by Marler and Boudreau (2017), which argue that leveraging HR analytics and AI can improve hiring decisions, optimize talent management, and ultimately boost organizational performance. The high significance of innovation in recruitment and retention strategies can be attributed to the importance of acquiring and retaining the right talent in a competitive market, which is vital for both productivity and performance. Furthermore, Wright and McMahan (2011) emphasized that HR practices focused on innovation are key drivers in improving organizational agility and performance.

A comparative analysis with prior empirical studies provides a critical lens through which to interpret the current findings more robustly. For instance, while this study confirms the significant impact of HR analytics and AI-driven innovations on workforce productivity and organizational performance consistent with (Marler & Boudreau, 2017) the contrasting non-significance of digital technology adoption, training, and flexible work arrangements suggests the influence of contextual variables unique to MSMEs and startups in Indonesia. Unlike large enterprises in developed economies, smaller firms in emerging markets often lack strategic frameworks, infrastructure readiness, or digital maturity, limiting the transformative potential of such interventions. These discrepancies underscore the necessity of contextual adaptation when applying global HRM innovations at the local level. Comparative studies thus reinforce the notion that successful digital transformation in HR is not only a matter of adopting best practices but also of aligning those practices with firm-specific capabilities, cultural readiness, and structural conditions (Bondarouk & Brewster, 2016). This perspective encourages future research to go beyond one-size-fits-all models by integrating contextual variables into the analysis of HRM effectiveness across different organizational settings.

The non-significant findings regarding digital technology adoption, digital training and development, and work flexibility within MSMEs and startups reflect deeper structural and contextual challenges that hinder their positive impact on workforce productivity and organizational performance. These challenges include inadequate technological infrastructure and limited financial and human resources, which constrain the ability of these enterprises to fully implement and utilize digital tools and flexible work arrangements effectively. Additionally, many MSMEs face a cultural resistance to change, rooted in long-established traditional practices and managerial mindsets that slow the integration of new technologies and innovative work models. The lack of strategic alignment between digital initiatives and broader organizational goals further diminishes their potential benefits. Collectively, these factors demonstrate that mere adoption of digital technologies or flexible work policies is insufficient; rather, successful transformation requires robust infrastructure, skilled personnel, adaptive organizational culture, and cohesive strategies to translate these efforts into tangible improvements in productivity and performance.

Although this study's findings contribute valuable insights, it is important to critically synthesize the conflicting results found in prior research regarding the impact of digital training on organizational performance. Some studies report positive effects due to skill enhancement and increased employee engagement, while others, including the present study, observe limited or non-significant outcomes. These discrepancies can be attributed to several contextual factors, such as differences in organizational readiness, the quality and relevance of training content, and the degree of alignment between training objectives and broader strategic goals. In smaller enterprises like MSMEs and startups, constraints including limited technological infrastructure, insufficient managerial capacity, and resource scarcity can hinder effective implementation and integration of training outcomes, thus delaying observable benefits at the organizational level. Moreover, the time lag between training delivery and performance improvements means short-term assessments may underestimate the true impact. Reconciling these divergent findings requires acknowledging these moderating variables and adopting a more nuanced, context-sensitive approach when evaluating digital training effectiveness.

The empirical results indicate that among the five independent variables examined, only Innovation in Recruitment and Retention, along with the Utilization of HR Analytics and Artificial Intelligence, exert statistically significant positive effects on both workforce productivity and overall organizational performance. This finding underscores the critical role of strategic innovation and data-driven decision-making in enhancing human resource outcomes within MSMEs and startups in Indonesia's digital economy. Conversely, the lack of significant impact from Digital Technology Adoption, Digital Training and Development, and Work Flexibility suggests that these factors may have more complex or context-dependent influences that are not captured by a straightforward positive relationship. It is possible that the effectiveness of these practices varies depending on organizational readiness, sector-specific demands, or implementation quality. Additionally, external factors such as resource constraints, employee digital literacy, or cultural resistance could moderate their impact. These nuanced results highlight the need for a more differentiated understanding of how digital HRM practices translate into measurable performance gains in emerging business contexts.

To provide a deeper understanding of the non-significant findings related to digital training and work flexibility, it is important to consider several contextual and practical factors. First, the ineffectiveness of digital training might stem from implementation challenges such as insufficient customization of training programs to fit the specific needs of MSMEs and startups, which often have limited resources and diverse employee skill levels. Additionally, a lack of strong managerial support or inadequate communication about the benefits of such training may reduce employee motivation and engagement, thereby diminishing its impact. Regarding work flexibility, its benefits may be contingent on the company's digital maturity and organizational culture. In less digitally mature organizations, flexible work arrangements may not be supported by the necessary infrastructure or managerial trust, limiting their positive effects on productivity and performance. Furthermore, the unique operational demands and resource constraints of MSMEs and startups may influence how these digital HRM practices translate into performance outcomes. These factors highlight the need for a more nuanced approach in applying digital HRM strategies, emphasizing the importance of contextual alignment and readiness for digital transformation to realize tangible benefits.

Theoretically, these results can be interpreted through the lens of the Resource-Based View (RBV) and the Dynamic Capabilities framework. The RBV posits that firms gain a competitive advantage by effectively managing valuable, rare, inimitable, and non-substitutable resources (Barney, 1991). The strong impact of HR analytics and innovative recruitment/retention practices supports this view, as these digital HR capabilities can be seen as strategic resources that contribute directly to value creation and sustainable competitive advantage. Furthermore, the Dynamic Capabilities perspective, as initially conceptualized by Teece et al. (1997), underscores an organization's critical ability to sense opportunities and threats, seize them by mobilizing resources, and reconfigure its asset base to sustain competitive advantage in volatile environments. This theoretical lens is particularly relevant in understanding the mixed empirical results observed in this study. The significant impact of innovation and analytics on organizational performance reflects firms' effective deployment of dynamic capabilities specifically, their capacity to integrate data-driven insights and foster innovative talent management practices that enable continuous adaptation and renewal of HRM systems. These capabilities allow organizations to align their internal

competencies with shifting external demands, thus enhancing overall performance outcomes. In contrast, the insignificant effects of digital adoption and work flexibility may highlight a lag in the maturation or strategic embedding of these capabilities within the firms' dynamic capability framework. According to Teece (2007), dynamic capabilities must be deeply integrated and contextually tailored to an organization's unique environment to generate value; mere presence of digital tools or flexible work arrangements does not guarantee impact unless they are purposefully configured and aligned with strategic objectives. Therefore, the findings suggest that while digital adoption and work flexibility represent potential capabilities, their underutilization or lack of integration reflects a gap in firms' dynamic capabilities to reconfigure HRM practices in response to environmental changes. This underscores the necessity for firms not only to adopt technological and flexible work innovations but also to develop the managerial and cognitive routines that enable effective orchestration of these resources in line with dynamic capabilities theory.

Therefore, this study not only confirms prior empirical findings but also offers theoretical insights by illustrating that digital HRM strategies must be considered within the broader organizational and environmental contexts. It also invites further research into the mechanisms through which digital training, flexibility, and technology adoption translate into performance especially in resource-constrained settings. Future studies should consider including qualitative data (e.g. employee interviews or case studies) to unpack internal organizational processes and contextual barriers that hinder the effectiveness of certain digital strategies. Exploring moderating variables such as leadership readiness, digital maturity, or cultural alignment may provide a deeper understanding of why some digital HRM initiatives succeed while others do not in MSMEs and startup ecosystems.

Many Micro, Small, and Medium Enterprises (MSMEs) in Indonesia, particularly in Central Java, have limited human resource (HR) capabilities, focusing primarily on production activities rather than a comprehensive approach to business management, which includes finance, marketing, and product quality aspects (Radyanto & Hayati, 2020). As the shift toward digitalization intensifies especially following the acceleration caused by the COVID-19 pandemic there is a growing need for MSMEs to adopt digital HRM practices that enable them to manage talent effectively in the industry 4.0 era (Prasetio et al., 2023). These practices include investing in digital tools, leveraging cloud-based applications, and training employees to efficiently operate digital platforms (Abduh et al., 2024). One significant development in this area is the introduction and standardization of Human Resource Information Systems (HRIS), which can enhance HR management processes by improving accuracy, efficiency, and strategic alignment with technological advancements (Wibawa et al., 2018). The integration of MSMEs into broader e-commerce ecosystems, often facilitated by cooperatives and digital intermediaries, has also shown promise in improving sales and overall business management. This transition, however, necessitates substantial investment in employee training to build the digital literacy needed for managing online platforms, customer interactions, and data-driven inventory systems (Sakti et al., 2024). Consequently, MSMEs and startups must prioritize building digital competencies within their workforce to ensure effective technology adoption (Prasetio et al., 2023). The implementation of standardized HRIS not only streamlines HR functions such as payroll, performance monitoring, and recruitment but also enhances organizational responsiveness and scalability[5]. Furthermore, fostering a workplace culture rooted in innovation and knowledge sharing is critical for driving both performance and market share in a competitive environment (Piliang et al., 2025). Looking ahead, the future of HRM in local enterprises will increasingly rely on agile, innovation-driven models that embrace continuous learning, strategic talent development, and digital transformation as core pillars of sustainable growth.

The strategic implementation of human resource management in the era of the digital economy must consider sector-specific dynamics to ensure relevance and applicability. In the culinary sector, which dominates MSMEs in regions such as Makassar City, the transition to digital platforms remains a critical challenge due to limited digital adoption (Abduh et al., 2024). Meanwhile, the maritime industry in the Riau Islands is hindered by centralized decision-making, limited human capital, and underdeveloped digital infrastructure (Pratiwi et al., 2024). In contrast, tourism-based MSMEs have demonstrated their importance in generating employment and local income (Hermawati & Suci, 2019), but still face constraints related to funding, upskilling, and operational efficiency (Andriansyah et al., 2024). Agro-based

enterprises are further pressured by regional trade agreements such as the ASEAN Economic Community (AEC), demanding strategic readiness and competitive human resources (Sarma et al., 2022). Entrepreneurial marketing is one approach that has proven effective in boosting market responsiveness and adaptability (Lisa et al., 2025). Additionally, manufacturing MSMEs, while contributing significantly to economic output, are also environmental polluters. The adoption of circular economy principles is increasingly vital to align HR strategies with sustainability goals (Nur et al., 2024). These sectoral comparisons highlight the importance of adaptive, innovation-driven, and context-specific HR strategies to navigate the complex challenges of the digital economy.

MSMEs in Indonesia often exhibit resistance to change, particularly in adopting new digital practices. This inertia stems from a lack of digital literacy and knowledge sharing, which hampers the effective implementation of flexible work strategies (Noerchoidah et al., 2025). The reliance on centralized decision-making within MSMEs limits the autonomy and flexibility needed for remote work arrangements. This structure can stifle innovation and responsiveness, making it difficult to adapt to flexible work models (Pratiwi et al., 2024). Slow regulatory reforms and bureaucratic hurdles further complicate the adoption of flexible work strategies. These constraints can delay the implementation of necessary changes and reduce the effectiveness of such strategies (Farliana et al., 2023). Limited technological infrastructure is a significant barrier. Many MSMEs lack access to reliable internet and advanced digital tools, which are essential for effective remote work and digital transformation (Hamjen et al., 2023). There is a notable gap in the development of human resources, particularly in digital skills. Without adequate training and development, employees struggle to adapt to new work models, reducing the effectiveness o flexible work strategies (Prasetio et al., 2023b). Access to finance remains a critical challenge for MSMEs. Financial constraints limit their ability to invest in necessary technologies and training programs that support flexible work arrangements (Said & Soi, 2025).

To complement the quantitative findings, incorporating qualitative insights provides a richer understanding of how HR analytics and recruitment innovations influence MSME performance. For instance, a qualitative study in the foodservice industry in Jakarta revealed that critical HRM practices including HR planning, recruitment and selection, training and development, performance appraisals, and compensation significantly enhance MSME business functionality and performance (Nurianna et al., 2023). Similarly, research in the maritime industry of the Riau Islands identified that Digital Human Resource Management (DHRM) systems improve efficiency and streamline recruitment processes, despite challenges such as limited human resources and inadequate technological infrastructure (Pratiwi et al., 2024). Additionally, in-depth interviews with HR professionals from various industries in Indonesia emphasized the crucial role of HR competencies, such as strategic thinking, leadership, and communication skills, in improving employee engagement, productivity, and profitability (Kosasih et al., 2024). These qualitative insights enrich the quantitative results by illustrating the practical mechanisms through which innovative HR practices and data-driven approaches translate into tangible performance improvements within MSMEs.

This study acknowledges the need to move beyond merely confirming existing theories and instead engage in a critical exploration of context-specific factors. While the non-significant findings for Digital Technology Adoption, Digital Training and Development, and Work Flexibility may initially seem to contradict the prevailing discourse on digital HRM effectiveness, they actually reveal the complexity of digital transformation in the MSME and startup environment. These organizations often operate under constraints such as limited capital, lack of specialized digital expertise, and inadequate digital infrastructure, which may hinder the successful implementation of technology-driven HR strategies. Importantly, the study's findings suggest that the mere adoption of digital tools is insufficient without simultaneous investments in capacity building, cultural readiness, and strategic alignment dimensions that were not directly measured in this study but could be explored in future qualitative or mixed-method research. In this regard, the study makes a valuable contribution by highlighting that the success of digital HRM practices is not universal, but instead is highly contingent upon organizational preparedness and context-specific enablers. Future research should incorporate additional variables such as leadership support, employee digital literacy, and organizational learning capacity to better explain variations in outcomes and develop a more comprehensive model of digital HRM effectiveness.

6. Conclusions

6.1. Theoretical implication

This study, conducted between January and February 2025 in Indonesia, contributes theoretically by examining how various dimensions of digital transformation in Human Resource Management (HRM) affect organizational performance and workforce productivity within Micro, Small, and Medium Enterprises (MSMEs) and startups. The findings offer a nuanced contribution to the literature on digital HRM by distinguishing which aspects of digital transformation yield tangible business outcomes. Contrary to prevailing assumptions, the study found that Digital Technology Adoption, Digital Training and Development, and Work Flexibility and Hybrid Management did not significantly influence either outcome variable. These findings suggest a boundary condition in digital HRM theory: not all digital interventions uniformly enhance performance. In contrast, the study confirmed that Innovation in Recruitment and Retention and the Utilization of HR Analytics and AI significantly improve both workforce productivity and organizational performance. This supports and extends Resource-Based View (RBV) and Dynamic Capability Theory by emphasizing the strategic value of tailored, analytics-driven, and innovation-focused HR practices. Specifically, the results point to the importance of contextual and content-specific digital strategies over general digitalization efforts, contributing to a more differentiated theoretical understanding of digital transformation in HRM.

6.2. Managerial implication

From a managerial perspective, the findings offer targeted, actionable insights for HR leaders and business owners in MSMEs and startups seeking performance gains through digital transformation. Rather than investing broadly in technology, organizations should concentrate on strategic innovations in recruitment and retention that are low-cost, scalable, and personalized. For instance, managers can implement AI-driven candidate screening tools that reduce bias and improve hiring speed, or use chatbots to streamline initial applicant engagement. Retention can be enhanced through continuous feedback platforms and data-informed career pathing, which provide real-time insight into employee satisfaction and development needs. Regarding HR Analytics and AI, rather than merely collecting HR data, firms should embed analytics into decision cycles by training HR staff in data literacy and linking analytics dashboards to performance KPIs such as turnover rate, time-to-productivity, and employee lifetime value. Managers should also consider forming cross-functional digital HR task forces to oversee the responsible use of AI in HR decisions, ensuring compliance and ethical standards. In terms of training programs, traditional online modules should be replaced with adaptive learning systems that personalize content based on employee skill gaps and learning styles, ideally linked to immediate work outcomes. Meanwhile, given the limited impact of work flexibility on productivity and performance, managers should avoid adopting hybrid models wholesale; instead, they should conduct pilot programs, assess job-role fit, and develop clear accountability frameworks before scaling. Ultimately, managers must take a selective, evidence-based approach to digital HR strategy, implementing only those interventions that are aligned with the organization's specific human capital needs and operational realities.

6.3. Suggestion

The findings of this study highlight the strategic value of adopting innovative human resource (HR) practices within Indonesia's MSMEs and startup sectors. Notably, enhancing recruitment and retention mechanisms, supported by the use of HR analytics and artificial intelligence (AI), has been shown to significantly improve workforce productivity and overall business performance. These practices can help smaller enterprises respond more effectively to changing market demands and talent expectations.

For business owners and managers, it is advisable to begin with low-cost, scalable digital tools that support informed HR decision-making. Even basic HR analytics systems can provide valuable insights into employee engagement, performance trends, and turnover risks. Additionally, fostering a supportive work

culture through flexible role structures, ongoing skill development, and recognition programs can increase employee loyalty, even when financial incentives are limited.

On a broader level, policy interventions should be aligned with the operational realities of local enterprises. This includes providing MSMEs with access to affordable HR technology solutions, training programs focused on digital and data literacy, and partnerships that connect smaller firms with larger organizations or industry mentors. Such measures can bridge capability gaps while promoting inclusive innovation across business types and regions.

To support Indonesia's transition toward a digitally integrated economy, the government and relevant stakeholders should prioritize scalable and cost-effective solutions for MSMEs. These may include subsidized HR tech platforms tailored for small businesses, national campaigns to raise awareness on digital HR practices, and incentives for enterprises adopting data-driven workforce strategies. Aligning these initiatives with Indonesia's digital economy roadmap will not only enhance competitiveness but also ensure sustainable employment growth, especially within underrepresented and rural areas.

While the adoption of digital tools and flexible work arrangements is recognized as essential, their impact within MSMEs and startups may vary depending on organizational maturity, resource availability, and industry context. A step-by-step implementation strategy, accompanied by practical guidelines and technical support, can ensure that such initiatives deliver tangible benefits without overburdening smaller firms. By aligning HR innovation with enterprise capacity and policy support, MSMEs and startups can enhance their human capital advantage and build a more adaptive, sustainable business model for the future.

It is important to acknowledge the unique constraints faced by low-resource enterprises, especially micro and rural-based businesses with limited financial and technological infrastructure. For these entities, recommendations should emphasize highly accessible and cost-efficient solutions, such as mobile-based HR platforms, cloud-based freemium software, and open-source HRM tools. Government support plays a crucial role in enabling this transformation through targeted subsidies, micro-grants, and public-private partnerships that lower the cost barriers to HR technology adoption. Additionally, digital training support tailored for non-technical users delivered through community-based learning centers or mobile outreach units can significantly boost the digital readiness of enterprise owners and HR staff. By addressing these specific limitations, the recommendations can become more inclusive and applicable across the entire spectrum of MSME capacities, ensuring that even the most resource-constrained businesses benefit from the shift toward strategic and technology-enabled human capital management.

In addition to addressing immediate challenges faced by MSMEs, it is crucial to consider the scalability and sustainability of HR innovation strategies as these enterprises grow and evolve. Strategies that work effectively for micro and small businesses must be adaptable to the changing organizational complexities and resource availability of medium-sized or larger enterprises. Scalable HR technologies, such as modular platforms that can expand functionalities over time and flexible digital training programs that evolve with workforce needs, are essential to ensure long-term viability. Moreover, embedding continuous improvement mechanisms and fostering a culture of innovation can help enterprises maintain agility during growth phases. By proactively integrating scalability into HR technology adoption and management practices, MSMEs can build resilient human capital systems that support sustainable development, competitive advantage, and operational excellence throughout their growth trajectory. This strategic foresight enhances the relevance and impact of the recommendations across different stages of enterprise maturity.

6.4. Research limitations

There are several limitations to this study that warrant careful consideration. First, the data was collected exclusively from MSMEs and startups in Indonesia, which may limit the external validity and generalizability of the findings to other countries, organizational types, or industries with different institutional or technological environments. Second, the study employed a cross-sectional, questionnaire-based data collection method, which, while efficient, is inherently subject to common

method bias and self-reporting inaccuracies. Respondents may have provided socially desirable answers or may have lacked full awareness of organizational strategies and outcomes, potentially skewing the results. Third, the sample may suffer from selection bias, as participation was voluntary and may have disproportionately attracted firms with a stronger digital orientation or interest in HR innovation, thereby limiting representativeness. Fourth, the study did not employ longitudinal data, which constrains the ability to infer causal relationships or observe dynamic changes in HRM practices and performance outcomes over time. Finally, while the research focused on specific digital HR strategies, it did not control for other potentially influential variables such as leadership style, organizational culture, or external market conditions, all of which could play critical roles in mediating or moderating the observed relationships. Future research is encouraged to address these limitations by employing mixed-methods designs, including qualitative interviews and longitudinal data collection, to provide a more nuanced and temporally grounded understanding of digital HRM in evolving business environments.

6.5. Future research

Future research could extend the current findings by incorporating additional variables that may serve as critical mediators or moderators in the relationship between digital HRM practices and organizational outcomes. For instance, employee engagement could be explored as a key psychological mechanism through which digital HR initiatives translate into productivity and performance gains. Similarly, digital leadership the ability of leaders to guide organizations through digital change, may significantly influence the effectiveness of technology-driven HR strategies, particularly in settings with low digital maturity. The inclusion of organizational culture as a contextual factor is also essential, as it shapes employee receptivity to innovation and can either enable or hinder the success of HR digitalization efforts. Future studies should also consider conducting industry-specific analyses, as the impact of digital HR strategies may differ substantially between sectors such as manufacturing, services, or creative industries, due to varying work processes, regulatory requirements, and technological demands. Moreover, longitudinal research designs would offer greater insight into the temporal dynamics and sustainability of digital HRM impacts, allowing scholars to capture delayed or compounding effects that cross-sectional designs cannot reveal. Employing mixed-method approaches, combining quantitative surveys with qualitative interviews or case studies, could also enrich the understanding of how and why certain digital HR strategies succeed or fail in different organizational contexts.

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Author contributions

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Data availability statement

Data are available at a reasonable request from the corresponding author.

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